Will the AI build-out pay for itself?

The one-page version of *The Macroeconomic Limits of AI Compute*, version 28 (31 October 2026)

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The news, and the story behind it. On 29–30 October AI stocks lost well over $1 trillion of value: Alphabet about $500B on Thursday, technology stocks about $1T on Friday (Nasdaq −2.9%, chip stocks −7%, Intel −29%); the Nasdaq-100 lost about $2T in October. Alphabet, Amazon and Meta reported negative free cash flow, and Meta and Alphabet are selling up to $180B of new shares to fund the spending. As one strategist put it, the market is "increasingly questioning whether shareholders will earn an adequate return on the enormous capital being deployed." That is exactly what this paper counts — and its answer is below. Counted in today's dollars, the hole on flat spending is about $2T, the same size as what the market marked down in October; if the builders keep raising spending, as they say they will, it is nearly twice that.

The question. This year Microsoft, Alphabet, Amazon, Meta and Oracle will spend about $780 billion on AI data centers. Musk's SpaceX/xAI, CoreWeave, Nebius and others add about $140 billion more. Will the money come back?

How we count. A Fermi estimate: many small factors, each bounded and sourced, with the uncertainty carried to the result. All 53 inputs are public.

The answer in five lines

- **The cost.** The buildings and chips cost about $890 billion a year to own by 2030 — wear, running costs and a return on the money — even if spending stays flat.
- **The revenue.** AI revenue is about $179 billion this year. To cover the cost by 2030 it must grow 57% a year, every year. Counted sector by sector it grows about 33%.
- **The gap.** About $420 billion a year short in 2030. Flat spending is the optimistic case: every company pointed 2027 spending up.
- **The wait.** Counting the early losses and the cost of the money, no spending path earns an ordinary return by 2035. Even a one-third cut turns a yearly profit only from 2032.
- **What would prove it wrong.** AI revenue above $440 billion in calendar 2028.

New in this version. **Every builder counted.** The rent that SpaceX/xAI, CoreWeave and the others earn from AI labs and clouds was always in the revenue we counted; their buildings and chips were not in the cost. Now they are. That one correction moves the 2030 gap from about $260 billion (the five largest alone) to about $420 billion. Builders that publish no figure yet (Meta's Hyperion, Crusoe, Lambda, the Gulf's Humain and Stargate UAE and others) are listed with what will bring each in; each would widen the gap.

Also new. *Inflation*, counted once: after it, flat spending is losing about 6.5% a year in 2035. *Tax*: the new write-off delays tax rather than cancelling it, so tax widens the gap. *The October earnings*: Anthropic's first filed numbers matched the paper's estimate.

Why it slows. Time saved is not money saved; checking AI's work takes time; much of the value goes to buyers and integrators; and the grid and local consent hold projects back.

What it means. Not a crash, and not the end of AI. The likely path is write-downs and a spending retreat in 2027–29, then a cheaper, more distributed build. This week moved the timing earlier, not the destination. The sign to watch: the first builder to guide 2027 spending down.

Where I was wrong. Earlier versions counted the other builders' rent but not their cost. It is corrected and listed, as every error will be.

Read more.

Paper, version 28: https://doi.org/[v28DOI] All versions: https://doi.org/10.5281/zenodo.23406475

Deck (19 slides): on the Zenodo record Model, inputs and code: https://github.com/desfitz/ai-compute-limits

What changed: https://github.com/desfitz/ai-compute-limits/blob/main/CHANGES.md First open question: https://github.com/desfitz/ai-compute-limits/issues/1

SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7620359 Comments: desmond.fitzpatrick@nvara.ai

Data Democracy: https://datademocracy.substack.com/p/the-macroeconomic-limits-of-ai-compute